

Mohamed Helmy

Flutter Developer | Software Engineer

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PROFILE

Flutter Engineer with 2+ years of experience building scalable, secure, and high-performance mobile & cross-platform applications, about creating innovative mobile applications that solve real problems. Specialized in Dart, Flutter, and modern Mobile applications development, AI integrations, performance optimization, and secure mobile systems.

SKILLS

Languages	Dart · Java · Kotlin · Python · C++ · C · SQL
Frameworks	Flutter · Android SDK · Google Maps API · Stripe SDK · Dio · HTTP
State Management	Bloc · Cubit · Provider · Async Programming · Multithreading
Architecture	Clean Architecture · MVVM · Repository Pattern · SOLID · OOP · Design Patterns
Backend & Cloud	REST APIs · Firebase Auth · Firestore · Realtime DB · Firebase App Distribution
Database	Hive · MySQL · Oracle · Secure Storage
AI / ML	Machine Learning · CNN · Computer Vision · AI Model Integration · NLP
DevOps & Tools	Git · GitHub Actions · CI/CD · Postman · Android Studio · VS Code
Testing	Unit Testing · Widget Testing · Performance Profiling · Background Services
Mobile Security	AES Encryption · Secure Storage · Firebase Auth · Role-Based Access Control

PROFESSIONAL EXPERIENCE

Flutter Developer | Gravigol (Traineeship)

Dec 2025 – Apr 2026 | Cairo, Egypt

Delivered cross-platform enterprise mobile applications for 4 major regional clients across telecom, aviation, and banking sectors.

- **STC:** Built feature-rich enterprise frontend using Provider/Bloc state management and RESTful API integration; eliminated UI state inconsistencies through strict MVVM layer separation, improving stability by ~35%.
- **SITA:** Built a subscription management app with dynamic content rendering and automated financial notification workflows, reducing manual billing follow-ups by ~70%.
- **ABK (Al Ahli Bank of Kuwait):** Improved screen rendering performance by 40% through widget tree restructuring and lazy loading; modernized internal banking UI/UX to enhance staff efficiency.
- **ADCB (Abu Dhabi Commercial Bank):** Shipped a full-scale banking app with 60fps native-like performance and reliable background data synchronization, improving client transaction reliability by ~35%.

Software Engineer – Military Service | Egyptian Armed Forces (MTC)

Dec 2024 – Dec 2025 | Cairo, Egypt

Engineered a secure Flutter system to digitize military educational logistics, lecture tracking, and attendance — fully replacing a paper-based workflow.

- Developed a Flutter application that automated attendance and reporting workflows, reducing manual data-entry time by approximately 80% and eliminating transcription errors.
- Architected an offline-first sync engine using Hive local storage, ensuring zero data loss in restricted or low-connectivity network environments.
- Automated multi-step reporting pipelines, cutting report generation from several hours to under 5 minutes while improving data accuracy for senior staff.

- Implemented AES-based secure storage and role-based access control for handling sensitive military data, meeting internal security compliance standards.

Flutter Developer | Freelancer (Self-Employed)

Dec 2024 (part time) – Present | Remote

- Design and deliver cross-platform Flutter applications using Clean Architecture, MVVM, and Bloc/Cubit state management for diverse client requirements.
- Integrate REST APIs with JSON parsing, Firebase Auth, Firestore, and third-party SDKs including Stripe payment processing.
- Apply performance profiling and optimization techniques to maintain smooth 60fps UI rendering and low memory usage across target devices.
- Implement secure storage, background services, async programming, and Android lifecycle best practices for production-grade reliability.

PROJECTS

Muadh – معاذ – AI-Powered Content Protection App

Jan 2026 – May 2026

Technologies: Flutter · Dart · Kotlin · ONNX Runtime · Google ML Kit · Hive · Android VpnService · AccessibilityService · DeviceAdmin API

- Architected and shipped an AI-powered Android content protection platform that automatically detects and removes explicit media entirely on-device, with zero cloud dependency.
- Engineered a 3-model AI ensemble (NSFW Classifier, Face Detector, and Skin Analyzer) with confidence-based scoring and early-exit optimization, reducing processing overhead by ~25% while maintaining detection accuracy.
- Integrated Android AccessibilityService to monitor on-screen content across applications in real time, providing configurable alerts and policy enforcement for explicit text detection.
- Engineered a real-time media monitoring pipeline using Android FileObserver to instantly analyze newly created images and videos, enabling continuous protection without manual scans.
- Implemented Android Device Admin security controls with encrypted PIN-based authorization, preventing unauthorized app removal and protection bypass attempts.
- Developed a custom DNS-level filtering engine using Android VpnService to block explicit websites across all browsers and applications, with no configuration required from the user.
- Delivered a modular Clean Architecture with Riverpod state management, Hive local persistence, automated service recovery on device reboot, and a real-time monitoring dashboard with per-source deletion analytics.

APKPure : [muadh APK for Android Download](#)

App Release And Documentation: [Release Muadh v1.0.0](#) ·

ScanCare – AI Medical Diagnosis App (Graduation Project)

Dec 2023 – Jun 2024

Technologies: Flutter · Python · CNN · Firebase · Bloc · Google Maps API · Stripe

- Developed an AI-powered medical app that analyzes medical images using Convolutional Neural Networks to detect health conditions with high diagnostic accuracy.
- Integrated 5 AI models, automating 70% of diagnostic tasks and significantly reducing average assessment time.
- Implemented Google Maps API for nearby clinic discovery, Stripe for secure online payments, and encrypted patient-provider messaging.
- Built on Clean Architecture with Bloc state management, achieving a maintainable and fully testable codebase.

Demo: [My Graduation Project](#)

E-Commerce Mobile App

Dec 2023 – Jan 2024

Technologies: Flutter · Firebase · MVVM · REST APIs · Stripe

- Full-featured platform with product browsing, cart management, favorites system, order history, and Stripe payment integration.
- Optimized list performance using pagination and lazy loading, maintaining smooth 60fps scrolling across low-end devices.

GitHub repo link : [MooHelmy/Ecommerce_app](#)

Bookly – Book Discovery App

Dec 2023 – Feb 2024

Technologies: Flutter · Bloc · REST APIs · MVVM

- Book search and reading app consuming public APIs; implemented response caching to reduce API calls by ~60% and improve load speed.

GitHub repo link : [MooHelmy/Bookly_app](#)

IoT Smart Home System

May – Jun 2024

Technologies: Arduino · Python · Firebase · DHT11 · Sensors

- Remote-control IoT dashboard for home appliances with real-time sensor monitoring: temperature, humidity, fire detection, and gas leakage alerts.
- Integrated Python-based voice recognition connected to Firebase Realtime Database for hands-free smart home automation.
- GitHub repo link : [MohaMdSalah2003/iot_project](#)

EDUCATION

Bachelor of Computer Science

2020 – 2024

Benha University – Faculty of Computers and Artificial Intelligence - Computer Science major | GPA: 3.3 |

Graduation Project: Excellent

Internships & Certifications

Android Developer — ITCRC	Jul – Oct 2022	Benha, Egypt
Mobile Dev using Flutter — ITCRC	Jul – Oct 2023	Benha, Egypt
Flutter Developer — Information Technology Institute (ITI)	Jan – Apr 2023	Menoufia, Egypt

Community & Content

Technical Content Creator – YouTube [@eng_mo723](#)

- Created 100+ educational videos covering Flutter, software engineering, embedded systems, Assembly programming, and Google Cloud.
- Built an engaged technical audience with 50,000+ total views across multiple curated playlists.

Soft Skills

Communication | Teamwork | Adaptability | Problem Solving | Time Management | Leadership